

DEV_IMPL_GUIDE - SUB-WF-TERMINATE-01-FLOW-WIK

1. Purpose

This guide explains how to implement and test the WIK termination BPMN as the flow executes at runtime.

Use it with:

- DD_SUB-WF-TERMINATE-01-FLOW-WIK-v1.0.md
- DD_SUB-WF-TERMINATE-01-Subscription_Termination-v1.0.md
- DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0.md
- DD_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md

This guide is specific to the BPMN process file:

sub-terminate-wik.bpmn

2. Runtime Actors

Runtime component	Role in this flow
subscription-workflow-api pod	Receives WIK termination request and calls the shared operation framework.
SUB-LM service/module	Provides authoritative subscription summary and status mutation APIs.
Drools	Evaluates rules.subscription.terminate.wik.
Camunda	Runs process sub-terminate-wik.
Worker apps	Execute external task topics from the BPMN. One app can handle many topics.
BIL-01	Computes TERMINATION_INTENT settlement.
FUL-05	Revokes service.
FUL-07/HomePass release	Releases HomePass slot or records durable stub request.
Message correlators	Correlate fulfillment messages back to Camunda.
OSR-RMA-01 EQP	Creates pickup work order from SubscriptionTerminated.
NOT-01	Sends customer notification after success.

3. Before The Flow Can Run

Confirm these objects exist before deploying or testing the BPMN:

Object	Required value
BPMN process key	sub-terminate-wik
Operation config	(operator_code=WIK, operation_kind=TERMINATE) -> sub-terminate-wik
Termination config	WIK row in subscription_termination_config
Drools package	rules.subscription.terminate.wik
Worker subscriptions	Required worker topics from the WIK DD
Message correlators	fulfillment-completed, fulfillment-failed
BIL-01 intent	TERMINATION_INTENT
FUL-05	Service revocation available
HomePass release	FUL-07 or approved durable stub

4. Step 1 - Caller Sends Request

Public endpoint:

```
POST /api/subscriptions/sub_01HZ_KAMAU_FIBER/terminate HTTP/1.1
Authorization: Bearer <jwt-or-service-token>
Idempotency-Key: term-sub_01HZ_KAMAU-20260720T143000-cc007
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "terminationTrigger": "CUSTOMER_REQUESTED",
  "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
  "equipmentDisposition": "PENDING_COLLECTION",
  "customerNotes": "Moving to a non-covered area; no service available at new address",
  "initiatingWorkflowKind": "BO_UI",
  "initiatingWorkflowRef": "cc-ticket-2026-07-20-4421"
}
```

Actor user and role come from the authenticated user token. For this example:

```
{
  "initiatingActorUserId": "cc-agent-wik-007",
  "initiatingActorRole": "CALL_CENTER"
}
```

5. Step 2 - API Pod Validates Shape And Idempotency

The subscription-workflow-api pod validates:

Field	Validation
operatorCode	Required and must be WIK for this BPMN.
terminationTrigger	One of the four parent DD trigger values.
terminationReasonCode	Required.
equipmentDisposition	Optional; if supplied must be CUSTOMER_RETAINS, PENDING_COLLECTION, or WRITTEN_OFF.

Idempotency behavior:

- First request inserts the operation and starts BPMN.
- Same Idempotency-Key with same request hash returns the existing operation response.
- Same Idempotency-Key with a different request hash must return 409 IDEMPOTENCY_KEY_REUSED.

Implementation note:

Do not insert a new operation before checking existing key and request hash. Use a transaction or unique constraint on the idempotency key plus operator/operation scope so concurrent duplicate requests cannot create two operations.

6. Step 3 - API Loads Required Data

6.1 Subscription Summary

The API gets current subscription state from SUB-LM.

Same deployable:

```
SubscriptionReadService.getSummary(subscriptionId)
```

Split services:

```
GET /internal/subscriptions/sub_01HZ_KAMAU_FIBER/summary
Authorization: Bearer <subscription-workflow-service-token>
```

Expected response:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operatorCode": "WIK",
  "customerId": "cust_01HZ_KAMAU",
  "statusCode": "ACTIVE",
  "homepassId": "hp_01HZ_KAREN_017",
  "packageRef": "pkg_FIBER_HOME_50M",
  "activatedAt": "2024-01-20T10:00:00Z",
  "contractMinimumTermMonths": 12
}
```

Do not use cache for current subscription status. Termination is a terminal state-changing workflow and must use the authoritative SUB-LM read path.

6.2 Termination Config

Read from subscription workflow persistence:

```
SELECT *
FROM subscription_termination_config
WHERE operator_code = 'WIK';
```

Required sample:

```
{
  "operatorCode": "WIK",
  "coolingOffWindowDays": 7,
  "defaultEquipmentDisposition": "PENDING_COLLECTION",
  "dunningEquipmentDisposition": "PENDING_COLLECTION",
  "dunningAutoWriteOffThreshold": 10000.00,
  "dunningAutoWriteOffCurrency": "KES",
  "customerSelfServiceEnabled": true,
  "coolingOffSupervisorOverride": true,
  "churnSaveAttemptEnabled": false
}
```

6.3 Equipment Bindings

Load equipment through OSR/asset inventory approved API or client wrapper:

```
GET /internal/subscriptions/sub_01HZ_KAMAU_FIBER/equipment-bindings
Authorization: Bearer <subscription-workflow-service-token>
```

Expected:

```
[
  {
    "equipmentRequirement": "GPON_ONT",
    "equipmentRef": "ont_01HZ_KAMAU"
  },
  {
    "equipmentRequirement": "ATA",
    "equipmentRef": "ata_01HZ_KAMAU"
  }
]
```

Do not query OSR database tables directly from termination code.

7. Step 4 - API Handles In-Flight Operation Interruption

If another state-changing operation is in-flight for the subscription:

```
external-cancel-requested
```

is correlated to the active operation. The framework waits up to 60 seconds.

If cancellation is acknowledged, termination continues. If not, return:

```
409 Conflict

{
  "errorCode": "OPERATION_INTERRUPTION_FAILED",
  "activeOperationId": "subop_01HZ_KAMAU_PAUSE_PRIOR"
}
```

8. Step 5 - API Evaluates Drools Before Starting BPMN

Call Drools package:

```
rules.subscription.terminate.wik
```

Facts:

```
{
  "operatorCode": "WIK",
  "subscription": {
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "statusCode": "ACTIVE",
    "activatedAt": "2024-01-20T10:00:00Z"
  },
  "request": {
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "equipmentDisposition": "PENDING_COLLECTION"
  },
  "config": {
    "coolingOffWindowDays": 7,
    "defaultEquipmentDisposition": "PENDING_COLLECTION",
    "dunningEquipmentDisposition": "PENDING_COLLECTION",
  }
}
```

```

        "customerSelfServiceEnabled": true,
        "coolingOffSupervisorOverride": true,
        "churnSaveAttemptEnabled": false
    },
    "actor": {
        "userId": "cc-agent-wik-007",
        "role": "CALL_CENTER"
    },
    "runtime": {
        "now": "2026-07-20T14:30:00Z"
    }
}

```

Expected decision:

```

{
    "eligible": true,
    "coolingOffApplicable": false,
    "daysActiveInContract": 920,
    "requiresCoolingOffSupervisorDecision": false
}

```

If Drools rejects, return the mapped parent-DD error and do not start Camunda.

9. Drools Rules To Configure

9.1 Eligibility

```

rule "WIK termination rejects already terminated"
when
    $req: TerminateRequestFact()
    $sub: SubscriptionStateFact(subscriptionId == $req.subscriptionId, statusCode == "TERMINATED")
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ALREADY_TERMINATED")
    }
end

rule "WIK customer requested termination allowed roles"
when
    $req: TerminateRequestFact(
        terminationTrigger == "CUSTOMER_REQUESTED",
        eligible == null,
        initiatingActorRole not in ("CALL_CENTER", "CUSTOMER_SELF_SERVICE", "SUBSCRIPTION_OPERATOR")
    )
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ROLE_NOT_AUTHORIZED_FOR_TRIGGER")
    }
end

rule "WIK customer self service termination requires config"
when
    $req: TerminateRequestFact(initiatingActorRole == "CUSTOMER_SELF_SERVICE", eligible == null)
    $cfg: OperatorConfigFact(customerSelfServiceEnabled == false)
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("CUSTOMER_SELF_SERVICE_NOT_ENABLED")
    }
end

```

9.2 Cooling-Off

```

rule "WIK cooling off applies only inside configured window"
when
    $req: TerminateRequestFact(terminationTrigger == "CUSTOMER_REQUESTED")
    $sub: SubscriptionStateFact(activatedAt != null)
    $cfg: OperatorConfigFact()
    eval(java.time.Instant.parse($sub.getActivatedAt())
        .plusSeconds($cfg.getCoolingOffWindowDays() * 86400L)
        .isAfter(java.time.Instant.parse($req.getRuntimeNow())))
then
    modify($req) {
        setCoolingOffApplicable(true)
    }
end

```

9.3 Equipment Disposition

```
rule "WIK caller supplied equipment disposition wins"
when
    $req: TerminateRequestFact(equipmentDispositionRequested != null)
then
    modify($req) {
        setEquipmentDisposition($req.getEquipmentDispositionRequested())
    }
end

rule "WIK dunning write-off disposition"
salience 10
when
    $req: TerminateRequestFact(terminationTrigger == "DUNNING_TERMINATION", equipmentDispositionRequested == null)
    $intent: BillingIntentDecisionFact(writeOffMarker == true)
then
    modify($req) {
        setEquipmentDisposition("WRITTEN_OFF")
    }
end

rule "WIK default equipment disposition"
salience -100
when
    $req: TerminateRequestFact(equipmentDisposition == null)
    $cfg: OperatorConfigFact()
then
    modify($req) {
        setEquipmentDisposition($cfg.getDefaultEquipmentDisposition())
    }
end
```

10. Step 6 - API Starts Shared Operation

After validation, interruption handling, and Drools approval, the API asks the shared SUB-WF framework to create the operation and start Camunda.

Same deployable:

```
OperationStartService.start(StartOperationCommand command)
```

Split services:

```
POST /internal/subscription-operations/start
Authorization: Bearer <subscription-workflow-service-token>
Content-Type: application/json
```

The payload below is the internal command passed to the framework start contract. It is not the public terminate API request.

```
{
  "operatorCode": "WIK",
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operationKind": "TERMINATE",
  "idempotencyKey": "term-sub_01HZ_KAMAU-20260720T143000-cc007",
  "initiatingActorUserId": "cc-agent-wik-007",
  "initiatingActorRole": "CALL_CENTER",
  "initiatingWorkflowKind": "BO_UI",
  "initiatingWorkflowRef": "cc-ticket-2026-07-20-4421",
  "bpmnVariables": {
    "customerId": "cust_01HZ_KAMAU",
    "homepassId": "hp_01HZ_KAREN_017",
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "resolvedEquipmentDisposition": "PENDING_COLLECTION",
    "coolingOffApplicable": false,
    "requiresCoolingOffSupervisorDecision": false,
    "equipmentBindings": [
      {
        "equipmentRequirement": "GPON_ONT",
        "equipmentRef": "ont_01HZ_KAMAU"
      }
    ],
    "customerNotes": "Moving to a non-covered area; no service available at new address"
  }
}
```

The framework:

1. Inserts `subscription_operation`.

2. Resolves (WIK, TERMINATE) to sub-terminate-wik.
3. Starts Camunda process sub-terminate-wik.
4. Stores the Camunda process instance id against the operation.

11. Step 7 - BPMN Runs drools-decide

The BPMN should run drools-decide again as a guard because state may have changed between API validation and Camunda execution.

If rejected:

```
kafka-emit SubscriptionTerminationRejected
subscription-operation-finalize FAILED
```

12. Step 8 - BPMN Moves Subscription To Pending Termination

Service task topic:

```
sub-lm-put-pending-status
```

Worker input:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "toStatus": "PENDING_TERMINATION",
  "currentTransitionType": "TERMINATE",
  "currentTransitionReasonCode": "CUSTOMER_MOVED_NO_COVERAGE"
}
```

13. Step 9 - BPMN Sends Billing Intent

Service task topic:

```
bil01-emit-intent
```

Worker input:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intentKind": "TERMINATION_INTENT",
  "intentContext": {
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "priorStatus": "ACTIVE",
    "coolingOffApplicable": false,
    "daysActiveInContract": 920,
    "contractMinimumTermMonths": 12,
    "equipmentDisposition": "PENDING_COLLECTION"
  },
  "cycleWindow": {
    "cycleStart": "2026-07-01",
    "cycleEnd": "2026-07-31"
  }
}
```

Expected output:

```
{
  "intentDecisionId": "bidec_01HZ_KAMAU_TERM",
  "settlement": {
    "lineItems": [
      {
        "kind": "PRORATED_REFUND",
        "amount": -1200.00,
        "currency": "KES"
      }
    ],
    "totals": {
      "net": -1200.00,
      "currency": "KES"
    },
    "refundAmount": 1200.00,
    "writeOffMarker": false
  }
}
```

14. Step 10 - BPMN Resolves Equipment Disposition

After billing, BPMN runs drools-decide again:

```
{
  "kjarPackage": "rules.subscription.terminate.wik",
  "requestedDecisions": [
    "resolve-equipment-disposition"
  ],
  "factsJson": "{caller disposition, config defaults, billing writeOffMarker}"
}
```

Expected:

```
{
  "equipmentDisposition": "PENDING_COLLECTION"
}
```

15. Step 11 - BPMN Handles Cooling-Off User Task

If supervisor decision is required, create user task:

```
cooling-off-override-decision
```

Candidate group:

```
COOLING_OFF_SUPERVISORS
```

Boundary timer:

```
PT24H
```

If supervisor rejects, emit SubscriptionTerminationRejected and finalize failed.

16. Step 12 - BPMN Calls FUL-05 To Revoke Service

Service task topic:

```
ful-call-terminate
```

Worker input:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intent": "TERMINATE_SERVICE",
  "reasonCategory": "CUSTOMER_REQUESTED",
  "equipmentDisposition": "PENDING_COLLECTION",
  "equipmentBindings": [
    {
      "equipmentRequirement": "GPON_ONT",
      "equipmentRef": "ont_01HZ_KAMAU"
    }
  ],
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_TERM:terminate"
}
```

The BPMN waits for:

```
fulfillment-completed
fulfillment-failed
```

17. Step 13 - Fulfillment Message Correlator Wakes BPMN

FUL-05 publishes or calls back with:

```
{
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_TERM:terminate",
  "terminalState": "FULLY_REVOKED",
  "completedAt": "2026-07-20T14:30:08Z"
}
```

The message correlator calls Camunda:

```
POST /engine-rest/message
Content-Type: application/json

{
  "messageName": "fulfillment-completed",
  "correlationKeys": {
    "fulfillmentCorrelationId": {
```

```

        "value": "subop_01HZ_KAMAU_TERM:terminate",
        "type": "String"
    }
}
}

```

18. Step 14 - BPMN Releases HomePass

Service task topic:

```
ful-call-release-homepass
```

Worker input:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "homepassId": "hp_01HZ_KAREN_017",
  "releaseReason": "SUBSCRIPTION_TERMINATED",
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_TERM:release-homepass"
}

```

If FUL-07 is not ready, use the approved durable stub. The stub must record the release request and expose retry/alert metrics.

19. Step 15 - BPMN Writes Master Fields

Service task topic:

```
sub-lm-put-master-fields
```

Worker input:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "fieldsJson": {
    "terminated_at": "2026-07-20T14:30:18Z",
    "termination_trigger": "CUSTOMER_REQUESTED",
    "termination_reason_code": "CUSTOMER_MOVED_NO_COVERAGE",
    "prior_status_at_termination": "ACTIVE",
    "last_status_changed_at": "2026-07-20T14:30:18Z"
  }
}

```

20. Step 16 - BPMN Commits Terminated State

Service task topic:

```
sub-lm-commit-state
```

Worker input:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "expectedFromStatus": "PENDING_TERMINATION",
  "toStatus": "TERMINATED"
}

```

Only after this succeeds can the BPMN emit SubscriptionTerminated.

21. Step 17 - BPMN Emits Event

Service task topic:

```
kafka-emit
```

Topic:

```
sophix.subscription.subscription.terminated
```

Payload:

```

{
  "eventType": "SubscriptionTerminated",
  "eventVersion": "1.0",
  "operationId": "subop_01HZ_KAMAU_TERM",
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "operatorCode": "WIK",
  "customerId": "cust_01HZ_KAMAU",
  "terminationTrigger": "CUSTOMER_REQUESTED",
}

```



```

    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "priorStatus": "ACTIVE",
    "newStatus": "TERMINATED",
    "terminatedAt": "2026-07-20T14:30:18Z",
    "equipmentDisposition": "PENDING_COLLECTION",
    "settlement": {
      "totals": {
        "net": -1200.00,
        "currency": "KES"
      },
      "refundAmount": 1200.00,
      "writeOffMarker": false
    }
  }
}

```

Write through the outbox. Do not block BPMN completion on Kafka broker availability.

22. Step 18 - BPMN Sends Notification

Service task topic:

```
notification-send
```

Worker input:

```

{
  "recipientKind": "customer",
  "recipientRef": "cust_01HZ_KAMAU",
  "notificationEventKind": "SUBSCRIPTION_TERMINATED",
  "eventContext": {
    "terminationTrigger": "CUSTOMER_REQUESTED",
    "terminationReasonCode": "CUSTOMER_MOVED_NO_COVERAGE",
    "terminatedAt": "2026-07-20T14:30:18Z",
    "equipmentDisposition": "PENDING_COLLECTION",
    "equipmentPickupExpected": true
  }
}

```

If notification fails, log it and continue. Do not compensate a successful termination because of notification failure.

23. Step 19 - BPMN Finalizes Operation

Service task topic:

```
subscription-operation-finalize
```

Worker input:

```

{
  "operationId": "subop_01HZ_KAMAU_TERM",
  "finalState": "COMPLETED"
}

```

24. Compensation Flow Details

Failure	BPMN compensation
Drools rejection	kafka-emit SubscriptionTerminationRejected, finalize failed.
Billing settlement failure	sub-lm-revert-to-prior, emit rejected, finalize failed.
FUL-05 failed before network change	sub-lm-revert-to-prior, emit rejected, finalize failed.
FUL-05 partial/unknown revoke	Manual recovery.
HomePass release failed	Retry or manual recovery unless durable stub accepted the request.
Master-field write or commit failed after network revoke	Manual recovery.
Notification failed	Log and continue.

25. Manual Smoke Test

1. Seed WIK subscription_operation_config and subscription_termination_config.
2. Deploy Drools package rules.subscription.terminate.wik.
3. Deploy BPMN sub-terminate-wik.

4. Start worker apps subscribed to required topics.
5. Create or identify an ACTIVE WIK subscription with equipment bindings.
6. Send termination API request with CUSTOMER_REQUESTED.
7. Confirm API returns 202 and an operation id.
8. Confirm Camunda process instance key is sub-terminate-wik.
9. Confirm subscription status becomes PENDING_TERMINATION.
10. Confirm BIL-01 receives TERMINATION_INTENT.
11. Correlate fulfillment-completed.
12. Confirm HomePass release worker records success or durable stub request.
13. Confirm subscription status becomes TERMINATED.
14. Confirm outbox contains SubscriptionTerminated.
15. Confirm OSR-RMA creates pickup work order for PENDING_COLLECTION.
16. Confirm operation final state is COMPLETED.

26. Common Mistakes

- Treating worker-topic names as shell commands.
- Reading current subscription status from cache.
- Forgetting to interrupt in-flight operations.
- Computing settlement inside SUB-WF instead of BIL-01.
- Removing the cooling-off supervisor user task from WIK.
- Committing TERMINATED before FUL-05 revokes service.
- Emitting SubscriptionTerminated before sub-lm-commit-state.
- Dropping HomePass release failures without a durable retry/stub.
- Failing the whole operation because NOT-01 notification failed.